

Deepak Mallampati

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Kent, OH (Open to Berlin relocation; STEM Master's / Blue Card eligible)

PROFESSIONAL SUMMARY

Full-stack production AI engineer with JavaScript/React, Python, FastAPI, and PostgreSQL depth, plus end-to-end AI workflow integration. Shipped Manga Lens solo to the Chrome Web Store with four AI vision providers under per-provider failure isolation across 29 sites. Built FastAPI + React multimodal platform (Dream Decoder) with structured prompt-transformation layer improving alignment ~30%. Production RAG and agentic workflows in regulated healthcare with measurable outcomes (~35% retrieval precision, >30% hallucination reduction, ~30% post-deploy defect reduction). End-to-end project ownership, business-to-technical translation, stakeholder docs. Master's in Computer Science, Kent State. Ready to ship marketing-automation integrations on day one.

CORE COMPETENCIES

JavaScript / React • Python / FastAPI • API Integration • PostgreSQL • AI Workflow Integration • End-to-End Ownership • Testing & Quality • Docker / CI-CD • Multi-Provider AI • Stakeholder Docs

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Full-stack AI, RAG, Agentic Workflows

USA - Healthcare Technology, AI Consulting

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval; recursive semantic chunking and transformer embeddings improved retrieval precision ~**35%**, reduced irrelevant retrieval >**30%**, response alignment >**90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules; response stability ~**25%**, hallucinations cut >**30%**.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects ~**30%**.
- Built preprocessing pipelines (Pandas, NumPy) over EHR extracts, appointment histories, support ticket logs; raised dataset reliability >**98%**, cut downstream model instability ~**40%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost); recall +**15-20%** on high-risk categories, precision held >**90%**.
- Wrote stakeholder-facing system-limitation docs and translated business requirements into technical specs for clinicians and operations staff.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas ERP

- Optimized **T-SQL stored procedures** on a compliance-sensitive ERP; query time **-20%**, retrieval latency **-25%**.
- Built **Jenkins CI/CD pipelines** for schema updates and stored-procedure deployments; deploy errors **-30%**, release cycle **-35 to -40%**.
- Designed performance dashboards with **SQL Server DMVs + Grafana**; incident recurrence **-25%**, root-cause resolution **-30%**.
- Supported RBAC and audit logging for financial modules.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ sessions; manual review cut **60-70%**, AI highlights aligned with human curation ~**85%**.
- Implemented **document Q&A with RAG**; hallucinations cut ~**30%**, contextual accuracy >**85%**.
- Deployed as **Flask** API with web interface for non-technical staff.

PROJECTS

Manga Lens - Multi-Provider AI Vision Chrome Extension (shipped, Chrome Web Store)

Full-stack browser extension built solo - Manifest V3, content scripts, service workers, captureVisibleTab. Integrates 4 AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Per-provider payload handling for worst-case failure isolation. 7-day translation cache, per-domain selector configs for 29 manga/webtoon sites, narrowed host permissions. Direct analog to marketing-automation tooling integrating multiple AI vendors.

TypeScript, Manifest V3, Service Workers, Claude / GPT-4o / GPT-4.1 / Ollama, WebP/JPEG payload routing

Dream Decoder - AI-Powered Multimodal Creative Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating Perplexity Sonar (interpretation) and Replicate (16:9 visuals). Intermediate structured prompt-transformation layers improved contextual alignment ~30% and first-pass image success ~25-30% over naive direct orchestration.

FastAPI, React + Vite + Tailwind, Perplexity Sonar, Replicate

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

5-stage multi-agent pipeline with schema-validated JSON contracts between agents. RAG-grounded CPT/ICD validation; ANN duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts

EDUCATION

Master's Degree - Kent State University

Aug 2023 - May 2025

Focus: Applied Machine Learning, NLP, Database Systems. STEM degree - eligible for German Blue Card.

SKILLS

Languages: JavaScript, TypeScript, Python, SQL (PostgreSQL, T-SQL, SQLite), PHP, C++ (Arduino)

Frontend: React, Vite, Tailwind, Chrome Extensions (Manifest V3), Service Workers

Backend: FastAPI, Flask, REST, API integration, Docker, structured logging

Data: PostgreSQL + pgvector, SQLite, Pandas, NumPy, ETL pipelines, data lineage

AI Workflow Integration: Claude, OpenAI, Anthropic, Ollama, multi-provider abstraction, RAG, agentic workflows, prompt engineering, evaluation pipelines

DevOps: Docker, Jenkins, CI/CD, Grafana, observability

Practice: Testing & code quality, business-to-technical translation, stakeholder docs, end-to-end project ownership